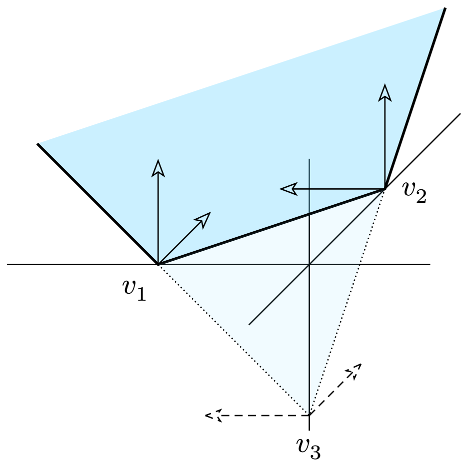
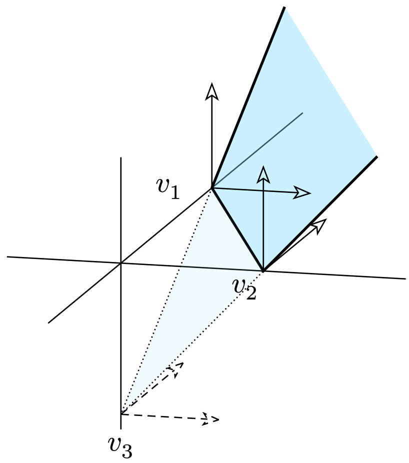
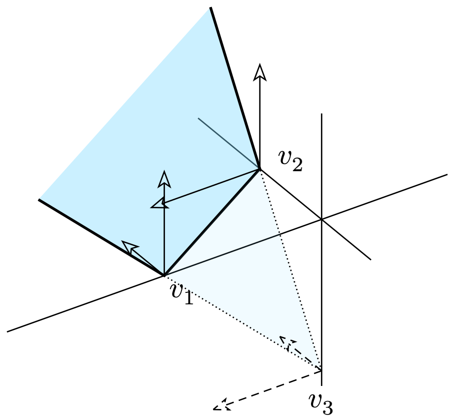
Basic solutions







Given a linear map $A : \mathbb{R}^n \to \mathbb{R}^m$ and a vector in its range $b \in \mathbb{R}^m$, form the linear system $Ax = b$.

A basis $\{e_1 ... e_n\}$ in the domain of A now gives many ways to express b as a combination of the image of the basis $\{Ae_i\}_i$:

$$b = \sum_{i \in B} \lambda_i A e_i$$

for many choices of m-sized $B \subset \{1...n\}$ and $\lambda_i, i \in B$. Not every choice for B makes b expressible as a combination of $\{Ae_i\}_{i \in B}$, but at least one does.

For each such choice of B, we can rewrite $b = A \sum_{i \in B} \lambda_i e_i$ and denote

$$x_B := \sum_{i \in B} \lambda_i e_i$$

so that $b = Ax_B$.

x_B is now well-defined for any choice B of (a subset of) basis vectors in the domain of A whose images form a basis of A's range. Such x_B are special solutions to the system $Ax = b$ and are called **basic** solutions.

Given B, those basis vectors e_i that form x_B are called **basic**, and the rest are called **non-basic**. In the sketches to the side, we see copies of the non-basic e_i attached to each basic solution e_i.

The solutions of $Ax = b$ are in general of the form

one particular solution of $Ax = b$	$+$	a solution of the homogeneous system $Ax = 0$

$$x_0 + x$$

Different choices for x_0 give rise to different representations of the solution space. Next, we'll see what happens when we put *basic* solutions for x_0.

Basic and non-basic variables

Given B and $x_B = \sum_{i \in B} \lambda_i e_i$, we get two natural decompositions of an arbitrary solution $x = \sum_i \lambda_i e_i$, $Ax = b$:

1. Its basic and its non-basic parts,

$$x = \underbrace{\sum_{i \in B} \mu_i e_i}_{\text{basic part}} + \underbrace{\sum_{i \notin B} \mu_i e_i}_{\text{non-basic part}}.$$

Keeping B fixed and varying x, we call $\lambda_i, i \in B$ the *basic* variables, and $\lambda_i, i \notin B$ – the non-basic variables.

2. The basic solution plus a direction,

$$x = x_B + (x - x_B).$$

We can see $x - x_B$ as the direction pointing from x_B to x; it lies in the kernel of A.

Keeping x_B fixed and varying x, we get different directions $x - x_B$ out of x_B.

TODO show how the two are related

Feasibility

The above discussion may looks as if it is about a general linear system, but it also presupposes a *choice of basis* is given, $\{e_1 ... e_n\}$. The definition of basic solutions, though, is invariant to scaling the basis vectors, i.e. it only depends on the choice of n independent *lines*.

In linear problems we also have inequality constraints, which in standard form take the shape of $x \geq 0$, meaning that all components of x *in the given basis* should be non-negative.

That is, the choice of basis determines a cone of positivity (in the 3D pictures above — an octant in the coordinate system), and solutions to $Ax = 0$ are called *feasible* only if they lie in that cone.

In particular, basic solutions, too, fall into two classes: basic feasible solutions and basic infeasible solutions. In the sketches above, v_1, v_2 are feasible and v_3 is infeasible.

Reduced cost

Given a B, we represent an arbitrary solution in the form $x = x_B + (x - x_B)$. Then an objective function given by $f(x) = \langle c, x \rangle$ acts on an arbitrary x by

$$f(x) = f(x_B + (x - x_B)) = \langle c, x_B \rangle + \langle c, x - x_B \rangle$$

When $x = x_B$, then $f(x) = \langle c, x_B \rangle$, i.e. the left term above is the value of f at that basic solution.

TODO show how $\langle c, x - x_B \rangle$ is computed.